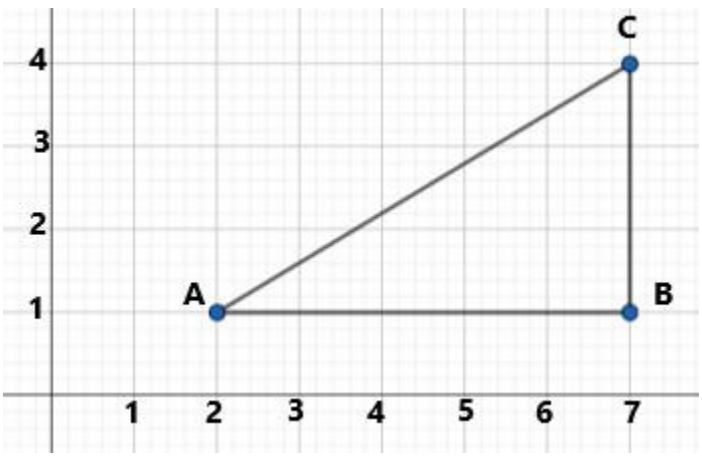
Programming Fundamentals II: Assignment 01

Due Date: Saturday September 28, 2024 at Midnight

Part 1. Polygons (20 Marks)

For this assignment, we are going to work with some basic functions on polygons in Java. We are going to represent polygons using ordered arrays of points. Each point is itself an array of doubles, where the doubles represent the coordinates of the point. A point may have varying numbers of coordinates, depending on how many dimensions the polygon is in. For example, a 2D polygon has points with 2 coordinates (x, y), a 3D polygon has points with 3 coordinates (x, y, z) and an n-dimensional polygon has points with n coordinates (x_0, x_1, x_2, … , x_n).

Below is a triangle, which we can consider 2-dimensional polygon consisting of 3 points.



For this 2D polygon, we have 3 points, {A, B, C} = { {2, 1}, {7, 1}, {7, 4} }.

The bounding box for this polygon can be represented as { {2, 1}, {7, 4} }.

The total length of the lines connecting the points, which we will refer to as the *length* of the polygon, is ~13.8 units.

Download the Ass01.java and Polygon.java files provided and make the following changes to the Polygon class:

a. **(2 Mark)** Complete the constructor and setter for the Polygon class. These methods should both accept an array of points and save it to the corresponding field on the Polygon class.
b. **(3 Marks)** Write the code for the calculateDistance() method. This method will accept two points as input, with each point being represented by a 1-dimensional array of doubles. You can calculate the distance between two points using Euclidean Distance.
c. **(3 Marks)** Complete the calculateLength() method. This method will determine the total length of the polygon if we were to connect each point to the next point in order. Note that our polygon is considered to be closed. This means that the last point should connect back to the first point.
d. **(3 Marks)** Write the calculateBoundingBox() method. This method should return an array of points that can be used to represent the maximum extents of the polygon. A bounding box for a shape can be determined using only two points. The first point will represent the minimum coordinates for the polygon in each dimension, while the second point represents the maximums.
e. **(3 Marks)** Write the code for the append() method. The append method of a Polygon should accept another Polygon as input. The method will then combine the Polygons together by adding all of the points from the input polygon to the array of the second polygon.
f. **(3 Marks)** Complete the checkDimensions() method. This method will accept an array of points as input. The method will then check that there is at least one point in the array, and that all of the points in the array exist in the same number of dimensions. (Eg: one array of points should not have points that are 2D as well as points that are 3D) Return true if the array passes the check.
g. **(3 Marks)** Finally, we are going to add some validation to our class. If a validation ever fails, an IllegalArgumentException should be thrown. Here are our validation rules:
    a. Anywhere that the points field on the Polygon class may be changed, we must check that all of the points in the array have the same number of dimensions.
    b. If we are to append two polygons together, we must ensure the two polygons are in the same number of dimensions.
    c. To calculate the distance between two points, the points must have the same number of dimensions.

Add the **date and your name and student number** at the top of the java file. Please submit your Polygon.java file to Nexus before the due date. **Late submissions will be not be accepted.**